

Course ID	Name	$T(x)$	$H(x)$	Average Score	Average Coursework
0	Electricity	5	3	2.7	55
1	Data Structures	8	3	3.8	59
2	Signals	12	4	1.5	40
3	SWE	16	2	3.6	50

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$$\mathbb{J}(\mathbf{S}_{given}) = (8 - 5) * 3 + (12 - 8) * 4 + (16 - 12) * 2$$

BestGapFactor(Signals)

$$\begin{aligned}
&= \Gamma \left(CreditHours * ((1 - \alpha) \cdot AverageCreditsLost + \alpha \cdot \Psi(AverageCourseworkMarksLost)) \right) \\
&= \Gamma \left(CreditHours * ((1 - \alpha) \cdot (MaxCredit - AvgCredit) + \alpha \cdot \Psi(MaxCoursework - AvgCoursework)) \right) \\
&= \Gamma \left(4 * ((1 - 0.5) \cdot (4 - 1.5) + 0.5 \cdot \Psi(60 - 40)) \right) = \Gamma \left(4 * (0.5 \cdot 2.5 + 0.5 \cdot \Psi(20)) \right) \\
&= \Gamma(4 * (1.25 + 1.15)) = \Gamma(4 * 2.4) = \Gamma(9.6) = [\mu - \sigma, \mu + \sigma] = [\mu - 2, \mu + 3]
\end{aligned}$$

BestGapFactor(DSA)

$$\begin{aligned}
&= \Gamma \left(CreditHours * ((1 - \alpha) \cdot AverageCreditsLost + \alpha \cdot \Psi(AverageCourseworkMarksLost)) \right) \\
&= \Gamma \left(3 * ((1 - 0.5) \cdot (4 - 3.8) + 0.5 \cdot \Psi(60 - 59)) \right) = \Gamma \left(3 * (0.5 \cdot 0.2 + 0.5 \cdot \Psi(1)) \right) \\
&= \Gamma(3 * (0.1 + 0)) = \Gamma(3 * 0.1) = \Gamma(0.3)
\end{aligned}$$

BestGapFactor(SWE)

$$\begin{aligned}
&= \Gamma \left(CreditHours * ((1 - \alpha) \cdot AverageCreditsLost + \alpha \cdot \Psi(AverageCourseworkMarksLost)) \right) \\
&= \Gamma \left(3 * ((1 - 0.5) \cdot (4 - 3.6) + 0.5 \cdot \Psi(60 - 50)) \right) = \Gamma \left(3 * (0.5 \cdot 0.4 + 0.5 \cdot \Psi(10)) \right) \\
&= \Gamma(3 * (0.2 + 0.15)) = \Gamma(3 * 0.35) = \Gamma(1.05)
\end{aligned}$$

$$\mathcal{L}(x, y) = \left(\text{BestGapFactor}(x) - (T(x) - T(y)) \right)^2$$

$$\mathbb{J}(\mathbf{S}) = \sum_{i=0}^{s-1} \sum_{j=0}^{C(\mathbf{S}_i)-1} \mathcal{L}(C(\mathbf{S}_i)_j, C(\mathbf{S}_i)_{j-1}), \quad C(\mathbf{S}_i) \text{ should be sorted by time}$$

$$\Psi(x) = \begin{cases} 0 & 0 \leq x < 3 \\ 0 & 3 \leq x < 7 \\ 0.3 & 7 \leq x < 11 \\ 0.7 & 11 \leq x < 14 \\ 1 & 15 \leq x < 18 \\ 1.3 & 17 \leq x < 22 \\ 1.7 & 20 \leq x < 25 \\ 2 & 24 \leq x < 29 \\ 2.3 & 27 \leq x < 33 \\ 2.7 & 30 \leq x < 36 \\ 3 & 33 \leq x \leq 40 \\ 4 & 40 < x \end{cases}$$

- $C(S_i)_j$ is the j th course taken by i th student.
- $\alpha \in [0, 1]$, is a weighing factor, higher α means more weight on current coursework rather than previous average.
- $T(C(S_i)_j)$ is the exam day of the j th course taken by the i th student,
 $C(S_i)_{-1}$ typically will just return the date of the first available timeslot. **TODO**
- $\mathcal{L}(x, y)$ is the loss function that takes two courses; x and y , and computes their loss
- $\mathbb{J}(S)$ is the cost function that computes the loss for each course taken by every student
- $\Gamma(x)$ is the normalization function that takes
the total number of available timeslots, should return
an expected gap length (from 1 to 7 inclusive).
- $H(x)$ is the number of credit hours of course x
- $F(x)$ is the average coursework of students in course x
- $SCORE$ is the maximum coursework score
- $\Psi(x)$ calculates the GPA lost given the lost marks in coursework